Complete Experiment Log

Every Experiment Run — Configuration, Results and Conclusion

Research Dossier Part 3 · Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems (PP-HFM)
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How to read this document. Experiments appear in the order they were run. Each entry states what was being tested, the configuration, the measured result, and the conclusion drawn. Conclusions are colour-coded: green where the experiment supported its hypothesis, red where it refuted it, amber where the outcome was inconclusive or mixed. Every number is traced to a file in `SIMULATORS/results/`. Section 6 lists experiments that have *not* been run, so the record is complete in both directions.

1 — Phase 4: Simulation Infrastructure

E1 NS-3 Packet-Level Network Simulation

Tests: whether the hierarchical topology actually reduces uplink traffic and round latency when payloads traverse a simulated network, rather than being computed analytically.

NS-3 v3.46.1 · 9 IoT devices, 2 edge servers, 1 cloud · device→edge 54 Mbps, edge→cloud 1 Gbps at 5 ms · LogDistance propagation, 5 Mbps background traffic · artifacts `ns3_results_{vitaldb,wear}.csv`, `ns3_phase5_{vitaldb,wear}.csv`

Quantity	VitalDB	WEAR
Round uplink time	33.97 s	—
Round downlink time	26.85 s	—
Round latency, flat → hierarchical	90.5 → 75.2 ms	104.5 → 81.8 ms
Latency reduction	**16.93 %**	**21.68 %**
Per-update reliability (2 hops)	**100 %**	**99.94 %**

Conclusion The edge tier reduces both traffic and round latency under packet-level measurement. Every network figure in this project comes from rounds where the payload actually traversed the topology — before this simulator was built, reported "NS-3 results" were the script's analytical fallback.

E2 CloudSim Plus Compute and Energy Simulation

Tests: energy cost per training round for centralised versus hierarchical federated learning.

CloudSim Plus v8.5.6 · IoT 300 MIPS / 512 MB / 1–3 W · edge 8000 MIPS / 16 GB / 50–200 W · GPU overhead modelled on edge servers only · `cloudsim_results_{vitaldb,wear}.csv`

Scenario	Total energy / round	Uplink
CentralizedFL	1266.25 J	73.85 MB
HFL_Proposed	**−72.3 % (VitalDB)**	see E10
HFL_Proposed	**−76.8 % (WEAR)**	see E10

Conclusion Energy saving of 72.3 % and 76.8 % against a 20 % target. **This is the energy figure to quote.** A second figure of 93.5 % exists from a linear bytes→Joules proxy, but under that model it is arithmetically identical to communication reduction and is not an independent energy measurement.

2 — Phase 5: Model, Privacy and Deployment

E3 Encoder Pretraining — Proxy vs Rigorous

Tests: the cost of the disjointness condition that makes parameter-efficient private training valid. If encoders are pretrained on data the clients also train on, ε does not hold end-to-end.

30 epochs, `--deterministic` · proxy mode = pooled FL train indices; rigorous mode = disjoint public split via `--public_data`

Mode	Effect on AUROC	Privacy validity
Proxy / oracle	baseline	**ε NOT valid end-to-end**
Rigorous public split	**−0.022**	**valid**

Conclusion A valid end-to-end guarantee costs **0.022 AUROC**. That gap is the quantity to report — not the proxy number alone. Determinism was verified: with the flag two runs of a seed are byte-identical; without it they diverge (epoch-1 loss 1.5699 vs 1.5713), and checkpoint selection can land on a different epoch.

E4 Baseline Suite B0–B5

Tests: where the proposed system sits against standard federated and centralised methods.

No frozen encoders, trained from scratch · `baselines/baselines_summary.csv`

Baseline	Test accuracy	Test macro-AUC	Note
B0 — Centralized	0.5787	0.8211	Upper reference for pooled data
B1 — Local-Only	0.4532	0.7511	Lower reference, no collaboration
B2 — FedAvg	0.6543	0.8795	Best non-private baseline
B3 — FedProx	0.6570	0.8747	No advantage over FedAvg here
B4 — DP-FedAvg (ε ≈ 1)	**0.2159**	**0.6443**	**Collapses.** The like-for-like DP reference
B5 — MOON	0.3934	0.7418	Underperforms on this task

B4 is the critical row: applying DP the conventional way at the same budget destroys the model. That is what makes the head-only result in E5 meaningful. ***These baselines are not a fair comparator for the ε=1 head-only number*** *— they train from scratch, while the proposed runs start from pretrained encoders. The only valid comparator is the ε=∞ arm of the same architecture.*

E5 Head-Only DP Fine-Tuning — the ε=1 Collapse Fix

Tests: whether freezing the encoders and privatising only the fusion head makes ε ≤ 1 usable, when end-to-end DP-SGD at the same budget produces a random model.

20 rounds, 5 edge rounds · δ=1e-5, C=1.0 · head = 560,645 / 2,157,766 params (26.0 %) · 3 seeds

Recipe	σ	Test AUC	Round time
End-to-end DP-SGD	11.3281	**0.562**	1046 s
Head-only DP fine-tuning	11.3281	**0.8742 ± 0.0072**	50 s

Conclusion **The noise was never the problem — σ is identical in both rows.** What changed is optimisation geometry: a near-linear probe over frozen pretrained features tolerates σ≈11.3, while training from scratch does not. Also ~20× faster per round. This finding reframes the standard "DP destroys utility" complaint. (Figures from the legacy corpus later withdrawn — the mechanism transferred and is confirmed on VitalDB by E8.)

E6 ε Sweep — End-to-End Recipe (superseded)

Tests: utility across budgets under the original end-to-end recipe.

Conclusion Characterises the *broken* recipe. Retained only as the "without our fix" arm. **Must not be presented as the proposed method** — see `epsilon_sweep/README`.

E7 ε Sweep — Rigorous Head Recipe

Tests: where the utility floor actually lies, now that ε=1 costs nothing measurable. Swept downward as well as up.

ε ∈ {0.5, 1, 2, 4, 8, ∞} · 10 rounds, head recipe · `epsilon_sweep_rigorous/vitaldb/`

ε	σ	AUROC	AUPRC
0.5	5.92	0.8287	0.5110
1	3.24	0.8319	0.5464
2	1.88	0.8323	0.5381
4	1.21	0.8356	0.5590
8	0.85	0.8342	0.5544
∞	—	**0.8158**	**0.5204**

Conclusion Utility is **flat from ε=0.5 to ε=8**, and every private arm *beats* the non-private one. DP is acting as a regulariser on this task. The utility floor is below ε=0.5, i.e. lower than the budget the system actually ships at.

E8 Objective 2b — Multi-Seed Accuracy Cost

Tests: the accuracy cost of ε=1 against a same-architecture non-private comparator, with enough seeds to put an interval on it.

30 rounds, head recipe, paired seeds · VitalDB n=5, WEAR n=15 (both pre-registered) · `o2b_multiseed/`

Dataset	Cost (pp)	95 % CI	Verdict
VitalDB	−0.92	[−2.82, +0.99]	**PASS**
WEAR	1.967	[1.760, 2.173]	**INCONCLUSIVE**

Conclusion VitalDB: **no measurable cost** — the point estimate is slightly negative. The defensible claim is "no measurable cost, bounded by the interval", not "zero cost". WEAR: the interval **straddles the 2 pp threshold**, so the objective can be neither passed nor failed. The seed count was pre-registered at 15 and **deliberately not extended** — extending it after seeing the result would be p-hacking.

E9 SADP — Adaptive Per-Group Privacy Budget

Tests: whether splitting the budget across groups beats a uniform allocation at equal total ε.

Conclusion — NEGATIVE It runs correctly and **underperforms uniform DP by construction**. The cross-group composition tax drives σ from 11.3 to **70**. Retained as a characterised negative result, not a claimed contribution. The transferable lesson: splitting a budget across groups is paid for at a punitive rate under composition.

E10 Gradient Compression and Uplink Measurement

Tests: actual bytes on the wire under top-*k* sparsification plus 8-bit quantisation with error feedback.

top-20 % sparsity × 8-bit → 20× nominal · error-feedback accumulator · `uplink_measured_*.json`

Measurement	Bytes / round
Full-model FP32 FedAvg	258,931,920
Head-only, uncompressed	67,277,400
Observed, post-fix	**16,821,300**

Defect found and fixed: the compressor took a fixed `k = numel × sparsity` of *every* tensor, so frozen encoders — whose delta is exactly zero — contributed 20 % of their elements as explicit zeros. Omitting all-zero tensors accounted for a 78.87 % reduction on its own.

Conclusion Uplink reduction **90.82 % (VitalDB)** and **96.08 % (WEAR)** against a 50 % target. The fix was proven not to alter learning: the dense path is **bit-identical** to the previous implementation over 5 rounds, and an isolated re-run moved AUC by 0.0001.

E11 ONNX Export and Inference Latency

Tests: end-to-end serving latency of the deployable artifact, including the conformal prediction step.

1000 timed runs after 100 warmup, batch 1 · path = preprocessing → ONNX forward → conformal set → serialise · `inference_latency/latency_summary.csv`

Precision	Provider	Model	Inference P95	Conformal	Total P95	QoS
FP32	CPU	10.64 MB	13.259	0.079	**16.363 ms**	**PASS**
FP32	GPU (CUDA)	10.64 MB	20.088	0.096	**22.476 ms**	**PASS**
INT8	CPU	6.55 MB	27.877	0.090	**31.028 ms**	**PASS**
INT8	GPU (CUDA)	6.55 MB	70.020	0.112	**73.150 ms**	**PASS**

*All times ms. The headline objective figure of **4.58 ms P95 (VitalDB)** and **1.22 ms (WEAR)** comes from the final objective run on the current corpora; the table above is the four-way precision/provider comparison.*

Conclusion Objective 3 passes in every configuration. Two findings matter beyond the pass. **First, the privacy apparatus is free at serving time**: conformal occupies 0.68 % of the path and the DP mechanism occupies none, because its noise is applied to gradients during training. The objection that privacy is too heavy for a latency-bound deployment is answered by measurement. **Second, INT8 is counterproductive** — see E12.

E12 INT8 Quantisation for Edge Deployment

Tests: the prescribed INT8 export, expected to be faster and smaller.

Conclusion — NEGATIVE INT8 is **1.90× slower on CPU and 3.25× slower on GPU** than FP32, and misses the 4 MB size target at 6.55 MB. Two independent causes were identified: dynamic quantisation only accelerates MatMul/Gemm, while this model's cost is dominated by convolutions and a recurrent path; and on CUDA the quantised ops have no GPU kernel and fall back to CPU — the runtime reports **3180 Memcpy nodes** inserted, which is exactly the 3.25× penalty. **Deploy the FP32 artifact.** If size later matters more than latency, the correct lever is static/QDQ quantisation or quantisation-aware training.

E13 Patient-Level vs Record-Level DP

Tests: the utility cost of stating the guarantee over patients rather than training records.

Both arms at matched budget · 5 seeds · `patient_dp/vitaldb/` · VitalDB: 48,712 windows / 6,257 cases, mean 7.79 windows per case, **100 % of cases contribute more than one window**

Conclusion The upgrade is priced at **1.77× σ**. Because every patient contributes more than one record, a record-level and a patient-level guarantee are different statements for *every individual in the dataset* — this is the premise the whole project rests on, and it is measured, not assumed.

E14 Membership Inference — Loss Threshold Sweep

Tests: the pre-registered hypothesis N-A, that record-level DP leaves patients exposed while patient-level DP defends them. Predicted ≈0.70 vs ≈0.50.

Patient-aggregated loss threshold · ε ∈ {1, 4, 8} · `mia_vitaldb*.csv`

ε	record-DP	patient-DP	gap
1	0.5461	0.5374	+0.0087
4	0.5512	0.5417	+0.0095
8	0.5459	0.5430	+0.0030

Conclusion Against a predicted gap of ≈0.20, the measured gap is under 0.01 at every budget. But a loss threshold is the weakest attack in the literature and its AUC is only a **lower bound** on leakage, so this null was **not reportable on its own**. That is what motivated E16.

E15 MIA Factorial — Leakage by Recipe

Tests: which training recipe leaks, independent of the privacy unit.

Recipe	Attack AUC	member / non-member loss
full-model, no DP	0.5122	1.81 / 2.31
head-only, no DP	**0.5847**	0.013 / 4.95
full-model, DP	0.5206	6.93 / 7.46

Conclusion — UNEXPECTED **Leakage is a property of the recipe, not the task.** Head-only fine-tuning — *our own recipe* — memorises members far harder than full-model training (p=1e−9, sd 0.0018 over 6 runs). This is a genuine caveat to the parameter-efficient DP fine-tuning literature and became claim N-A′ when N-A failed.

E16 LiRA — Likelihood-Ratio Attack at ε=1

Tests: N-A again, under the strongest available attack, with a **pre-registered gate**: the attack had to exceed 0.5847 on a model known to memorise, or its nulls would be inadmissible.

3 arms × 16 shadow models · per-patient IN/OUT Gaussians · identical splits (`--split_seed 7`) · 5 paired seeds · 600 members + 600 non-members

Arm	Best attack AUC (95 % CI)	Via	Difficulty floor (E17)	Margin
Positive control — head-only, no DP	**0.6277** [0.5905, 0.6649]	lira_offline	0.5626	**+0.0627**
Record-level DP	0.5477 [0.4876, 0.6079]	lira_offline	0.5252	+0.0226
Patient-level DP	0.5224 [0.5175, 0.5274]	loss threshold	0.5167	+0.0057

Attack variant	Paired Δ (record − patient)	95 % CI	p
loss threshold	+0.0138	[−0.0034, +0.0310]	0.090
lira_offline	+0.0325	[−0.0280, +0.0929]	0.210
lira_online	+0.0034	[−0.0059, +0.0128]	0.365
lira_online_gv	−0.0211	[−0.0487, +0.0065]	0.101

Conclusion — N-A REFUTED **Every CI includes zero.** The control cleared its gate (+0.0513 over plain thresholding), so the null is a statement about the models, not a failing attack. State it precisely: **the large predicted effect (Δ≈0.20) is refuted; a small effect is unresolved at n=5** (half-width ±0.060). Never write "there is no effect". Supporting diagnostic: on the patient arm LiRA *adds nothing* (−0.0071, plain loss wins) — the signature of a defence with no difficulty-confounded signal left.

E17 Difficulty Floor — Confound Control

Tests: whether the MIA evaluation split is difficulty-balanced. Ranks patients by shadow-model difficulty alone, **using no target model at all**. Shadow membership is randomised over the whole patient universe, so a balanced split must score 0.5.

Added 2026-08-20 · computed from shadow scores already on disk, no retraining · `mia_difficulty_floor.py`, `mia_difficulty_floor_vitaldb.csv`

Arm	Floor	Attack	Margin
head_nodp (control)	0.5626	0.6254	**+0.0627**
record-DP, ε=1	0.5252	0.5477	+0.0226
patient-DP, ε=1	0.5167	0.5224	+0.0057
record-DP, ε=16	0.5743	0.5465	**−0.0278**

Conclusion — METHODOLOGICAL Members are drawn from released train patients and non-members from released test patients, and **those pools differ in intrinsic difficulty** (mean phi 10.25 vs 7.71). An attack is therefore rewarded for detecting difficulty, not membership. **Every MIA AUC must be read against its floor, not against 0.5.** The control still clears its floor, so the audit remains valid — but the refutation of N-A is **firmer** than first reported, because less of the apparent leakage was real. Floors differ per arm because better-fitting models separate easy from hard patients more sharply.

E18 *k*-Scaling Law — the Headline Result

Tests: whether the cost of a patient-level guarantee is a fixed penalty or a function of how many records each patient contributes.

All 24 configurations: $k \in \{1,2,4,8\}$ × 3 seeds × 2 arms · per-step budget and units-per-step held constant · `k_scaling/vitaldb/k_scaling_vitaldb.csv`

k	Sample arm AUPRC	Patient arm AUPRC	Gap	95 % CI	p
1	0.5015	0.5067	−0.0052	[−0.0664, +0.0561]	0.75
2	0.5590	0.5084	**+0.0507**	[+0.0417, +0.0596]	0.0017
4	0.5924	0.5092	**+0.0832**	[+0.0694, +0.0971]	0.0015
8	0.5955	0.5089	**+0.0866**	[+0.0603, +0.1129]	0.0049

Conclusion — THE CORE FINDING The cost is **a curve in *k*, not a number**: zero at k=1, growing monotonically, saturating by k=8. **The experiment carries its own positive control**: at k=1 a patient *is* a record, so the two guarantees are identical and the arms must collapse — they do (p=0.75). The mechanism is **asymmetric and must be stated that way**: the patient arm is flat in k (0.4967–0.5176 throughout) while the sample arm rises (0.5015→0.5955). The widening gap is the sample arm improving, not the patient arm degrading — extra records per patient buy the patient arm nothing, because noise scales with the patient's total contribution.

Honest limits: AUROC shows the same monotone shape (+0.0006 → +0.0094 → +0.0133 → +0.0132) but **no *k* reaches significance**. Report AUPRC as the resolving metric and AUROC as directionally consistent but underpowered at n=3. Accuracy is pinned at 0.8005 in every patient row — an uncalibrated 0.5 threshold on imbalanced data, **do not quote it**.

E19 Modality Ablation — Does Fusion Help?

Tests: whether combining two genuinely paired modalities beats the better single modality.

8 epochs · VitalDB 5 seeds (34,044 train / 7,360 test), WEAR 2 seeds · `ablation_{vitaldb,wear}.json`

Arm	VitalDB AUROC	WEAR AUROC
A only (signals / inertial)	0.8283 ± 0.0062	0.9976
B only (clinical / video)	0.7680 ± 0.0033	0.9957
Fused	**0.8653 ± 0.0052**	0.9980
Gain over best unimodal	**+0.0370**	**+0.0004**

Conclusion — MIXED VitalDB: fusion beats the better modality by **5.9× the seed sd**. Gap #1 demonstrated. **WEAR cannot corroborate it** — its best single modality already reaches 0.9976, so the task has no headroom and the script's own verdict reads "CEILING". Reported as a finding rather than hidden: **on some IoT tasks one modality suffices, so multimodal FL is not universally motivated.**

Correction on record: at 2 seeds the gain read +0.0411 AUROC / +0.1062 AUPRC. At 5 seeds it is +0.0370 / +0.0671 — AUPRC fell **37 %**. The conclusion holds but the 2-seed figure was seed-lucky. **Quote only the 5-seed numbers.**

E20 Feature Attribution — What the Model Uses

Tests: which inputs drive predictions, and whether privacy changes that.

> Integrated gradients, midpoint rule, 64 steps, completeness 6.06 % · 20 runs across 5 recipes × 2 baselines, plus a separate seed-stability check

Recipe	Mask share of attribution
full_dp	28.2 %
patient-DP	22.0 %
sample-DP	21.6 %
full_nodp	19.9 %
head_nodp	15.8 %

> **Conclusion** **Differential privacy shifts the model's attention toward missingness.** Under noise the model leans more on *which measurements exist* than on their values, monotonically with how much DP is applied. Rank correlation stays ρ=0.95 head-only private vs not.
>
> **Evidence, not proof**, for the related claim N-7: masks carry 22.0 % of attribution mass over 5 seeds and `ART_MBP_observed` is 4.99 % ± 0.33, top-5 on 4 of 5 seeds. Say **"consistently among the strongest", not "#2"**. Needs the mask ablation to become proof.

E21 Objective Verification O1–O4

Tests: all quantified objectives, resolved from measured artifacts with no hardcoded values.

Objective	Target	VitalDB	WEAR
O1 Uplink reduction	≥ 50 %	**90.82 % PASS**	**96.08 % PASS**
O1b Round latency reduction	> 0 %	**16.93 % PASS**	**21.68 % PASS**
O2a Privacy budget (training + conformal)	ε ≤ 1	**0.9705 PASS**	**0.9656 PASS**
O2b Accuracy loss under DP	≤ 2 pp	**−0.92 pp PASS**	**1.967 pp INCONCLUSIVE**
O3a Inference latency P95	< 100 ms	**4.58 ms PASS**	**1.22 ms PASS**
O3b Successful updates	> 99 %	**100 % PASS**	**99.94 % PASS**
O4 Energy saving per round	≥ 20 %	**72.30 % PASS**	**76.84 % PASS**

*VitalDB 7/7 PASS. WEAR 6 PASS + 1 INCONCLUSIVE. O2a is a **composed** budget — training 0.7705 + conformal release 0.2000 — which is stricter than the common practice of reporting training only.*

3 — Phase 6: Component Ablation and Stress Testing

E22 Component Ablation A0–A6

Tests: which parts of the system actually contribute. Each arm removes one component from the full system.

> 3 seeds per arm · VitalDB · `phase6/vitaldb/ablation.csv` · **Note:** A1–A4 had never produced a result until four silent defects were fixed on 2026-08-16 — a dead `--local_epochs` argument, a uint8-hardcoded quantiser, a summary reading `test_*` from a CSV holding only `val_*`, and a 1e-4 verdict threshold.

Arm	Configuration	AUROC	AUPRC	ε	Uplink	Verdict
A0	Full system	0.8541 ± 0.0035	0.5960	0.7705	251.97 MB	Reference
A1	No C1 — flat tier (no edge)	0.8484 ± 0.0057	0.5911	0.7704	50.40 MB	Small utility cost; uplink not comparable (flat topology counts differently)
A2	No C2 — no DP	**0.8307 ± 0.0060**	**0.5136**	∞	251.97 MB	**Removing DP makes it WORSE** — DP regularises
A3	No C3 — dense (no top-*k*)	0.8553 ± 0.0015	0.6037	0.7705	**1259.75 MB**	+0.0012 AUROC for **5× the traffic**
A4	No C4 — float32 (no quantisation)	0.8543 ± 0.0036	0.5959	**403.14 MB**		+0.0002 AUROC for 1.6× the traffic
A5	No C5 — unimodal	**0.8283**	**0.5417**	—	—	Fusion contributes (see E19)
A6	+ ASWA aggregation	0.8542 ± 0.0037	0.6019	0.7705	251.97 MB	**+0.0005 — no contribution**

> **Conclusion** Three results matter. **(1) A2: removing DP makes the model worse** (0.8307 vs 0.8541), confirming E7 — DP is acting as a regulariser, not a tax, on this task. **(2) A3 and A4 justify the compressor**: dense transmission buys +0.0012 AUROC for 5× the traffic, and float32 buys +0.0002 for 1.6×. Both compression stages are earning their place. **(3) A6: ASWA contributes nothing** (+0.0005 AUROC, accuracy actually −0.0042). It was a leftover from a scrapped earlier design and has been **deleted**.

E23 NS-3 Stress Tests

Tests: behaviour as round count and packet error rate rise.

> Round sweeps r30 / r50 / r70 / r90 / r105 / r120 / r250, plus packet-error sweeps · both datasets · `ns3/stress*.csv`, 17 files

Conclusion The system holds across round counts up to 250. This sweep also exposed a **definition defect**: update reliability had been computed as `hop_success ^ n_rounds` — the probability that *all* rounds succeed — which decays with round count, so *any* system fails a ">99 %" target for a long enough run. Per-update reliability is 99.8001 %; the whole-run figure is still emitted under the distinct name `run_reliability_all_rounds_pct`.

4 — The Revival Gate

E24 LiRA at ε=16 — N-A's Last Chance

Tests: whether N-A failed because ε=1 crushed both arms to chance, leaving no room for a gap to appear. At ε=16 the noise is 13× lower (σ 11.33 → 0.85), so a membership signal should survive if one exists.

Run 2026-08-19/20, completed · 6 targets + 32 shadows, 3 seeds · `mia_epsilon_lira.py` · `lira_eps/vitaldb/eps16.0/`

Arm	Best attack	Floor	Margin
Record-level DP	0.5465 (loss)	0.5743	**−0.0278**
Patient-level DP	0.5630 (lira_offline)	0.5259	+0.0371

Attack variant	Paired Δ (record − patient)	95 % CI	Significant?	Reading
loss threshold	+0.0163	[+0.0123, +0.0204]	**yes, positive**	Predicted direction, but an eighth of the predicted size — and the record arm is below its own floor, so this is confound, not leakage
lira_offline	**−0.1788**	[−0.2553, −0.1024]	**yes, reversed**	Artifact: the record arm's offline score is 0.3842, far below chance, from over-correction on an asymmetric difficulty tail
lira_online	+0.0057	[−0.0137, +0.0252]	no	The principled two-sided variant. Chance.
lira_online_gv	**−0.0552**	[−0.0896, −0.0208]	**yes, reversed**	Also points the wrong way

Conclusion — N-A CLOSED PERMANENTLY **The four attacks contradict each other.** Two report a significant gap in the *opposite* direction to the prediction, one reports a significant gap an eighth of the predicted size, and the principled two-sided variant reports chance. Mutually inconsistent estimators are the signature of no underlying signal, with each estimator being driven by a different facet of the difficulty confound.

Decisive on its own terms: **the record-level arm sits 0.0278 BELOW its own difficulty floor**. It carries no membership signal at all, so there is nothing for a record-versus-patient gap to be made of. Loosening the budget sixteenfold changed the record arm's attack AUC by 0.0012 (0.5477 → 0.5465).

The "no room at ε=1" hypothesis is refuted. N-A did not fail for want of headroom; it failed because the effect is not there. This closes the question rather than leaving it open at n=5, and it does so with the difficulty floor as a witness. Do not re-run at higher budgets — the ceiling has been tested and the answer did not change.

5 — Not Yet Run

Listed so the record is complete in both directions.

Experiment	Why it matters
N-D — normalised patient clipping	The only remaining candidate for an *original mechanism* rather than a measurement. A patient's gradient is summed over k windows then clipped to a fixed C, so long recordings are systematically down-weighted; normalising by k or $\sqrt{k}$ should fix it. Low effort, high upside.
eICU-CRD external validation	Tests whether the k-law (E18) is structural or dataset-specific. Blocked on PhysioNet credentialing. The prediction must be frozen **before** access.
Conformal coverage validation	The coverage guarantee is implemented and its ε cost is composed into O2a, but empirical coverage has not been measured.
Mask-channel ablation	Would turn E20's attribution evidence into proof for N-7.
S3 — hierarchical, no DP	Completes the topology axis alongside A1 and A2.
Fig. 4 redraw from measured scores	The LiRA figure is still synthetic and captioned "Illustrative". The measured per-patient IN/OUT distributions now exist in `shadows/*/scores.npz`; redrawing turns a teaching diagram into evidence.
Multi-seed B4/B6/B7	Would put intervals on the baseline comparisons.

6 — Summary of Outcomes

#	Experiment	Outcome	One-line result
E1	NS-3 network simulation	**SUCCESS**	16.9–21.7 % round-latency reduction, packet-level
E2	CloudSim energy simulation	**SUCCESS**	72.3 % / 76.8 % energy saving
E3	Encoder pretraining, proxy vs rigorous	**PRICED**	A valid guarantee costs 0.022 AUROC
E4	Baselines B0–B5	**SUCCESS**	Conventional DP (B4) collapses to 0.6443 AUC
E5	Head-only DP fine-tuning	**SUCCESS**	0.562 → 0.8742 AUC at identical σ
E6	ε sweep, end-to-end recipe	**SUPERSEDED**	Characterises the broken recipe only
E7	ε sweep, rigorous recipe	**SUCCESS**	Utility flat ε=0.5→8; DP beats no-DP
E8	O2b multi-seed accuracy cost	**MIXED**	VitalDB −0.92 pp PASS; WEAR inconclusive
E9	SADP adaptive privacy	**NEGATIVE**	Composition tax drives σ 11.3 → 70
E10	Gradient compression	**SUCCESS**	90.8 % / 96.1 % uplink reduction
E11	Inference latency	**SUCCESS**	4.58 ms / 1.22 ms P95, >20× margin
E12	INT8 quantisation	**NEGATIVE**	1.9× slower CPU, 3.25× slower GPU
E13	Patient-level vs record-level DP	**MEASURED**	Unit upgrade priced at 1.77× σ
E14	MIA loss-threshold sweep	**NULL**	Gap < 0.01 at ε = 1, 4, 8
E15	MIA factorial by recipe	**UNEXPECTED**	Head-only memorises hardest: 0.5847 vs 0.5122
E16	LiRA at ε=1	**N-A REFUTED**	All four Δ CIs include zero; control passed
E17	Difficulty floor control	**METHODOLOGICAL**	Attacks must be read against 0.52–0.57, not 0.5
E18	***k*-scaling law**	**CORE RESULT**	Cost is a curve in k; zero at k=1 by construction
E19	Modality ablation	**MIXED**	VitalDB +0.0370; WEAR at ceiling, +0.0004
E20	Feature attribution	**SUCCESS**	DP shifts attention to missingness, 15.8→28.2 %
E21	Objective verification	**SUCCESS**	VitalDB 7/7; WEAR 6 + 1 inconclusive
E22	Component ablation A0–A6	**SUCCESS**	DP helps; compression justified; ASWA deleted
E23	NS-3 stress tests	**SUCCESS**	Holds to 250 rounds; found a definition defect
E24	LiRA at ε=16 (revival gate)	**N-A CLOSED**	Attacks mutually contradictory; record arm below its own floor

24 experiments run, all complete. 14 succeeded outright, 5 returned characterised negative results, 3 were mixed or inconclusive, 1 was superseded, and 1 priced a cost.
***Every negative and inconclusive outcome is reported rather than dropped** — that is what makes the positive results credible.*